

Michael Allen Heroux

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Experience

Senior Research Scientist, ParaTools, Inc. October 2024 – present. Lead of the [PESO Project](#). Lead architect of the [E4S software ecosystem](#) and the [HPCG Benchmark](#). Participate in national and international strategies for advancing the role of reusable scientific libraries and tools. Cultivating a strategy for advancing scientific software ecosystems as platforms for improving effective and efficient scientific application development.

Visiting Scholar, Hewlett Packard Enterprise. November 2024 – present. Participate in planning, strategy, and design of software capabilities for HPC-AI applied to scientific problems using HPE Cray systems.

Consulting Analyst, Hyperion Research. November 2024 – present. Write about current topics and trends in software for HPC-AI applied to scientific problems.

Professor Alumnus, Department of Computer Science, Saint John's University. May 2026 – present.

Scientist in Residence and Adjunct Faculty Member, Department of Computer Science, Saint John's University. September 1998 – 2026, Scientist in Residence 2004-present. Teach Computer Science Research Methodologies, Parallel Computing, and Software Engineering courses. Direct undergraduate research theses in parallel computing and related areas. Participate in curriculum development.

Senior Scientist, Retired, Center for Computing Research, Sandia National Laboratories. May 1998-October 2024. Principal member 1998-2004, Distinguished member 2004-2016, Senior Scientist 2016 – October 2024. Led efforts of national and international scope in HPC, including as Director of Software Technology for the US Exascale Computing Project. Conducted research and development of scientific algorithms and software. Participated in strategic planning for the Center for Computing Research. Participated on program and standards committees in areas of expertise. Lead architect of the [E4S software ecosystem](#), the [Trilinos libraries project](#), the [Mantevo](#) applications performance-modeling project, and the [HPCG Benchmark](#).

Group Leader, Scalable Computing, Algorithms and Capability Prototyping Groups, SGI/Cray Research. March 1995-May 1998. Led a team of specialists in scientific

computing. Directed activities and participated in development, porting and optimization of large-scale parallel applications for SGI/Cray systems. Participated in and led standardization efforts for scientific computing. Led efforts in development of new application capabilities. Provided applications analysis and requirements to future computer systems development including the Cray T3E, T90, J90, SV1 and SV2.

Numerical Analyst, CFD Group, Engineering Applications, Cray Research. September 1993-February 1995. Responsible for research and development of numerical methods for engineering applications in CFD, structural analysis, electronics and reservoir simulation. Worked with application developers on Cray vector multiprocessors and distributed memory machines. Particular areas of interest were the solution of sparse and dense linear systems, iterative methods, parallel algorithms and large-scale scientific computation. Served as consultant on numerical methods for Cray Research customers and application specialists.

Numerical Analyst, Mathematical Software Research Group, Cray Research. October 1988-September 1993. Conducted research and development of numerical linear algebra libraries. Served as consultant on numerical methods for Cray Research customers and application specialists. Developed libraries of high-performance software for Cray Research computer systems.

Education

Ph.D. Mathematics. May 1989, Colorado State University, Fort Collins, Colorado.

M.S. Mathematics. August 1986, Colorado State University, Fort Collins, Colorado.

B.A. Mathematics. December 1983, Saint John's University, Collegeville, Minnesota.

Professional Awards

- 2026 SIAM Activity Group on Supercomputing Career Prize, 2026.
- HPCWire 35 HPC Legends, 2025.
- Secretary of Energy Achievement Award, 2024.
- IEEE Technical Committee on Parallel Processing Outstanding Service and Contributions Award, 2023.
- Fellow, Society for Industrial and Applied Mathematics (SIAM), 2019.
- Senior Member, IEEE, 2018.
- Distinguished Member of the Association for Computing Machinery, October 2009.
- 2015 HPCWire "People to Watch 2015" selection.
- 2014 FLC Regional Technology Transfer Award for Mantevo.
- R&D 100 Award for Mantevo 1.0, 2013, project initiator and leader.
- Best Poster Award, SC11 Conference, November 2011.

- ASC Salutes Profile, NNSA/ASC profile, September 2007.
- R&D 100 Award for Trilinos 3.1, 2004, project initiator and leader.
- SC2004 HPC Software Challenge Award, 2004.
- Member of Cray Research Gordon Bell Prize Finalist Team, 1996.
- Sandia Employee Awards:
 - Mantevo Team R&D 100 Award and External Impact on HPC Co-Design, 2013.
 - Winning X-caliber proposal for the DARPA/UHPC Program, 2011.
 - Educating the next generation of computational scientists, 2010.
 - IAA Algorithms Team, 2009.
 - Organizing Next-generation Applications Workshop, 2008.
 - Xyce/Charon/Algorithms Team, 2008.
 - Supercomputing Architecture & Programming Environment Team, 2008.
 - Leading Trilinos 7.0 Release, 2006.
 - Leadership of Trilinos Project, 2004.
 - Xyce Development, 2004.
 - Efforts in Nanosciences Initiative, 2003.
 - Algorithms for Circuit Simulation, 2001.
 - Parallel Circuit Simulation Code, 2000.

Professional Leadership

- Initiated the creation of ACM new Transactions on AI for Science as co-chair of ACM New Publications committee, 2023 – 2025.
- PI of the Post-Exascale Software-Sustainability Organization (PESO) project, 2023 – present.
- Director of Software Technology, The US Department of Energy Exascale Computing Project, November 2017-2024.
- Member, Committee of Visitors for NSF Office of Advanced Cyberinfrastructure, 2022.
- Member, Advisory Committee for NSF Office of Advanced Cyberinfrastructure, 2018 – 2023, Chair of Reproducibility & Sustainability Working Group.
- Member, Molecular Sciences Software Institute (MolSSI) Scientific Advisory Board, 2019 – 2022.
- Member, SIAM Fellows Selection Committee, 2020 – 2021.
- Committee Member, Wilkinson Prize for Numerical Software, 2020.
- Member, SC Conference Steering Committee, 2019 – 2020.

- Committee Member, Sidney Fernbach Prize in High Performance Computing, 2018 – 2021.
- Committee Member, NISO Committee on Reproducibility and Badging, 2019 – 2020.
- Reproducibility advisor to the Conference Chair, Supercomputing 2019 Conference.
- Reproducibility Chair, Supercomputing 2018 Conference.
- Technical Papers Chair, Supercomputing 2017 Conference.
- Gordon Bell Prize Committee member, 2016 – 2019.
- Chair, SC16 Test of Time Award Committee.
- Scientific Libraries Lead, DOE Exascale Computing Project, 2016 – 2017.
- Chair of the NITRD, multi-agency workshop on Computational Science and Engineering Sustainability and Software Productivity (CSESSP) Challenges., October 15 – 16, 2015, Washington, DC.
- Principle architect and developer of the HPCG benchmark code.
- Editor-in-Chief, ACM Transactions on Mathematical Software, 2010 – 2017.
- Created the Replicated Computational Results review for ACM Transactions on Mathematical Software, 2013 – 2015.
- Member of SC Conference Test of Time Award Committee, 2014 – 2015.
- Applications Program Chair, SC13 Technical Program, 2013.
- Editor, SIAM Book Series on Software, Environments and Tools, 2012-present.
- Subject Area Editor, Journal on Parallel and Distributed Computing, 2011-2016.
- Lead writer of Software Section in the International Exascale Software Project (IESP), 2011.
- Chair of DOE Application readiness review for Titan 20PF computer system, 2010.
- Created Career and Junior Scientist Awards for SIAM SIAG/SC, 2009-2010.
- Led SIAG/SC committee to select Career/Junior Scientist winners, 2009-2010.
- Led SIAG/SC committee to select 2010-2011 officers, 2009.
- Wrote whitepaper for NSF on sustainable software engineering, 2009.
- Member, International Exascale Software Project (IESP), 2008-2013.
- Sandia rep, DOE/ASCR Breakthroughs Report, 2009.

- Sandia PI, The Exascale Software Center (ESC), 2010-2012.
- Sandia PI, The SciDAC-2 TOPS-2 project, 2005-2010.
- Sandia PI, The Extreme-scale Algorithms & Software Institute (EASI), 2009-2012.
- Sandia PI, Institute for Advanced Architectures & Algorithms, 2008-2010.
- Associate Editor for SIAM Journal on Scientific Computing, 2010-2016.
- Chair of the SIAM Supercomputing Special Interest Group, 2008-2009.
- Program Director for SIAM Supercomputing Special Interest Group, 2000-2003.
- Program Chair for 2004 SIAM Parallel Processing Conference.

Scientific Software Leadership

- **Extreme-scale Scientific Software Stack (E4S):**
 - I initiated the E4S effort as part of my role as Director of Software Technology for the US Exascale Computing Project. E4S is a first-of-a-kind curated portfolio of reusable scientific software targeting next-generation computing platforms (and more traditional HPC systems).
 - E4S drives quality improvement, accessibility, compatibility, and community engagement in the development and use of software for science.
 - The primary E4S website is <https://e4s.io>
- **HPCG Community Benchmark for HPC Systems:**
 - In 2013, I started a new benchmarking effort called HPCG, in collaboration with Jack Dongarra, at the request of DOE NNSA. I initiated the HPCG strategy to complement the LINPACK benchmark for the TOP 500 list.
 - I am the architect and implementer of the reference version of the benchmark code and have worked directly with community members and vendors on design and implementation features.
 - I organized a series of community meetings to build understanding and acceptance of the benchmark in the international community.
 - All major computer vendors have an optimized version of HPCG, which displays the critical features we want to probe for future systems.

- Since its release in 2017, HPCG has received considerable coverage in the HPC press and has been the subject of more than 1800 publications.

HPCG is officially part of the TOP500 benchmark suite since ISC2017.

- **Kokkos Performance Portability for Parallel Node Architectures:**

- I am the founder of the Kokkos project as it started within the Trilinos project.
- I shepherded the migration of Kokkos out of Trilinos as it was clear that the needs for parallel execution patterns over compile-time polymorphic arrays was far greater than just the Trilinos user community.

- **Trilinos Project:**

- I initiated the Trilinos project 21 years ago to produce a compatible collection of independently developed mathematical software tools.
- Trilinos is a 2004 R&D 100 award winning product.
- I have led the Trilinos project through several distinct transition phases: First, the expansion of project functionality beyond solvers; then the transition from a Sandia-centric project to one that includes non-Sandians as first-class developers; then the (still ongoing) transition to scalable manycore, accelerator and heterogeneous systems and now the expansion of the Trilinos ecosystem to include non-native packages.
- Trilinos has grown from the original 3 packages to 60 and represents the world's largest active mathematical software project.
- Trilinos has user communities worldwide. We host tutorial events and annual user group meetings in the US and Europe.
- Since its first release in 2003, Trilinos has been cited more than 6000 times.

Trilinos provides the collaboration and delivery framework for many Sandia activities including most ASC algorithms efforts, numerous LDRDs, Office of Science projects and CRADAs.

- **Extreme-scale Scientific Software Development Kit (xSDK):**

- With Lois Curfman McInnes, I initiated the creation of the xSDK, a collection of 20+ community math libraries.
- xSDK establishes a community approach to improving math library quality and accessibility.

- The primary xSDK website is <https://xsdk.info>

Professional Service

- Member of Publications Board of the Association for Computing Machinery (ACM), 2020 – present, Chair of the New Publications Committee, Member of AI Task Force, ongoing.
- Member of the Swiss National Supercomputing Center (CSCS) Scientific Advisory Board, ongoing.
- Chair of the Platform for Advanced Scientific Computing 2023 (PASC23) conference.
- Visiting committee member, CEA high-performance computing review, 2016.
- Advisory Board Member, NSF-funded FLAME project, U of Texas at Austin.
- Reviewer for NSF in computational science and scalable computing, 2003-2016.
- PhD committee, Daniel Sunderland, University of Utah, 2014-2016.
- PhD committee, Fan Ye, Maison de Simulation, Saclay-Paris, 2014-2016.
- PhD committee, France Boillod-Cerneax, University of Lille, Paris, 2014.
- PhD committee, Radu Popescu, Ecole Polytechnique Federale de Lausanne, 2012-2013.
- PhD committee, Sarah Knepper, Emory University, 2010-2011.
- PhD committee, Bryan Marker, University of Texas at Austin, 2010-2011.
- Referee for SIAM Journal of Scientific Computing, SIAM Review, ACM Transactions on Mathematical Software, IEEE Transactions on Parallel and Distributed Systems, 1999-present.

Professional Memberships

- Fellow, The Society for Industrial and Applied Mathematics.
- Distinguished Member, The Association for Computing Machinery.
- Senior Member, IEEE.

Community Contributions and Impact

- **BSSw Fellows program:**
 - I started the Better Scientific Software (BSSw) Fellows program in 2017, inspired by work at the Software Sustainability Institute in the UK.
 - Each year, we take into account diversity as a fundamental element in the selection process. The list of fellows shows that we have had good success in attracting and selecting a diverse group of fellows: <https://bssw.io/pages/meet-our-fellows>
- **Collegeville Workshop Series:**
 - I created the Collegeville workshop series on scientific software.
 - The series focus is on bringing academics, industry and lab scientists together to discussion and advance the understanding of how scientific software is developed and used
 - The workshop series is described here: <https://collegeville.github.io/Workshops/>
- **Research Software Science (RSS) thrust:**
 - I am cultivating and promoting the strategy of research software science, applying the scientific method to understanding how software is developed and used to do research.
 - I have added cognitive and social scientists to my research teams, to better understand how their skills and contributions can accelerate advancements in scientific software.
 - The core concepts of RSS are discussed here: https://digitalcommons.csbsju.edu/csci_pubs/36/
- **Leadership Scientific Software platform (LSSw):**
 - Started and lead the community platform for discussion of post-Exascale software organizational requirements.
- **Better Scientific Software web portal (BSSw.io)**
 - I initiated and established the BSSw.io project with Lois McInnes and created the GitHub-based architecture for the BSSw.io web portal. The project has established a formal editorial staff and review process. This basic GitHub architecture is still used today.

- **Numerical Linear Algebra:**
 - Underlapping for domain decomposition: I developed the original concept of using “underlapped” subgraphs and subdomains that permit the use of standard preconditioners for communication avoiding (s-step) Krylov solvers. This idea led to the first practical preconditioning strategy for CA iterative methods.
 - Complex linear systems solution methods: I developed (with David Day) new algorithms and spectral theory for solving complex-valued linear systems via equivalent real formulations. These formulations permit the use of commonly available real-valued math software for which there is no similar complex-valued version (which is very common). This work has been wide used, especially in the electromagnetics community.
- **Proxy applications for HPC co-design:**
 - I implemented the first miniapp (HPCCG) and led the Mantevo project from the beginning. With colleagues at Sandia I demonstrated the value of miniapps in co-design activities.
 - Every co-design effort across DOE uses miniapps (or more generally proxy apps) in the way that Mantevo does.
 - Mantevo miniapps are cited in more than 200 publications over the past four years.
 - Mantevo has expanded to be an international community project with contributions from 7 institutions outside of Sandia, and a community web portal at <http://www.mantevo.org>.
- **Resilience:**
 - I have developed a taxonomy for application-driven resilient computing models, including two new approaches that I pioneered:
 - **Local-failure-local-recovery (LFLR):** This approach promotes a recovery model whose cost and scope is proportional to scope of failure. LFLR has emerged as a promising next step for practical application resilience for cases where global checkpoint-restart is costly or infeasible.
 - **Selective Reliability:** This model permits application and library developers to declare data and compute regions to be more (or less) reliable than the default execution environment. This kind of selectivity enables the

development of new algorithms where the majority of data and computation are in low reliability mode, but some portion of data and computation are in high reliability mode, ensuring the resilience of application execution.

- I have developed and promoted an additional model, relaxed bulk synchronous parallel (rBSP), that some application teams have implemented as a way to mitigate performance variability on emerging systems.
- LFLR is integrated into ASC product R&D plans, and its scalable recovery has been demonstrated on more than 10,000 processes.

- **Scientific Productivity:**

- I have participated in the definition, scoping and strategic discussions for a productivity-focused approach to advancing computational science and engineering.
- I have participated in numerous workshops on productivity over the past two years and the writing of multiple DOE reports.
- I was the keynote speaker at an inter-agency workshop on productivity in August 2014 and an invited member for an SC'14 panel on scientific productivity.
- I led (with Lois McInnes, Argonne and David Moulton, LANL) the original IDEAS Project, the first DOE project funded on scientific productivity.
- I lead (with Lois McInnes as primary lead, Argonne) the IDEAS Project under the US Exascale Computing Project.

Publicly-available Software Products

- **E4S (e4s.io)** A curated software stack for delivering a comprehensive software portfolio to the HPC community. I am the founder and project lead.
- **HPCG Benchmark (hpcg-benchmark.org)** Official TOP500 benchmark along with LINPACK for ranking the performance of the top high performance computing systems. I am the benchmark designer and implementer of the reference code. ***HPCG is a 2021 R&D 100 Honorable Mention.***
- **Kokkos (kokkos.org)** Performance portability library and programming model. Project founder and designer of core strategy.
- **The Trilinos Project (trilinos.org)**: Open Source (LGPL/BSD), Initiated and lead the project, 2001-present. ***Trilinos is a 2004 R&D 100 winner and the world's largest open source computational science and engineering libraries project.*** It is a collection of nearly sixty open source software packages supported by a common software engineering infrastructure and community development model.
 - **Trilinos package development:** Each Trilinos package is a self-contained software product with its own scope of development. These are the packages I have designed and developed:
 - **JuliaPetra:** A Julia language design and implementation of the Petra data object model used by Trilinos.
 - **Epetra:** Principal designer and implementer. Epetra is the predecessor to Tpetra and is one of the two most popular scalable data class libraries on the planet (PETSc is the other). Epetra is used by thousands of application and library developers for constructing and using scalable sparse and dense linear algebra objects.
 - **AztecOO:** Principal designer and implementer. An object-oriented version of the popular Aztec linear solver library. AztecOO is the most widely used iterative solver package in Trilinos, used by thousands of people, providing the core linear solver capabilities for many Sandia and DOE applications.
 - **Tpetra and Kokkos:** Initial designer and developer; remain an algorithm designer and funding source.

- **Ifpack:** Principal designer and implementer. A collection of algebraic sparse preconditioners and smoothers. Widely used in Sandia and DOE applications.
- **Ifpack2:** Initial designer. Next-generation of Ifpack targeting scalable manycore architectures.
- **Amesos:** Principal designer. A package of interfaces to common direct sparse solvers. Widely used at Sandia and other DOE labs.
- **Amesos2:** Initial designer. Next-generation of Amesos targeting scalable manycore architectures.
- **Belos:** Designer. Follow-on to AztecOO as a collection of scalable, state-of-the-art iterative methods.
- **Komplex:** Principal designer and implementer. A package of solvers for complex-valued systems using equivalent real formulations.
- **Teuchos:** Designer and developer. The core services package in Trilinos. Widely used.
- **The Mantevo Project (mantevo.org):** Open Source (LGPL), Initiated and continue to lead the project, 2006-present. ***Mantevo is the first project to concretely define the concept of a miniapplication as a co-design vehicle for next generation applications and computer systems. It was a 2013 R&D 100 winner.*** Mantevo is a collection of more than 16 open-source, stand-alone miniapplications that serve as performance proxies for Sandia's large-scale applications.
 - **Mantevo package development:** Each Mantevo miniapplication is a self-contained software product. These are the packages that I have designed and developed:
 - **HPCCG:** Principal designer and developer. Performance proxy for a scalable finite-volume/finite-difference single physics PDE application. ***HPCCG has been used in dozens of performance studies for new system design. Rewritten many times using new programming languages and programming models.***
 - **MiniFE:** Designer and developer. Follow-on to HPCCG as a proxy for unstructured finite element single physics

applications. Used to prototype manycore algorithms and parallel pattern implementations that are now in production use in Trilinos. Used in numerous systems performance studies on mixed precision and hybrid MPI+threading programming environments.

- **Tramonto (software.sandia.gov/tramonto):** Open Source (LGPL), Lead scalable algorithms designer and developer, 2004-present. Tramonto is an open source application for modeling and simulation of inhomogeneous fluids using classical density functional theories. ***Tramonto has unique modeling capabilities for a wide variety of applications, including biophysics applications for new pharmaceuticals based on anti-microbial peptides.***
- **Aztec (www.cs.sandia.gov/CRF/aztec1.html):** Open Source (Special license), Lead developer, 1998-2000. ***Popular open source preconditioned iterative solver package.***
- **Sparse BLAS (math.nist.gov/spblas):** Open Source (no license), Lead designer, 1999-2002. The sparse BLAS are a *de facto* standard for sparse kernel computations.
- **BPKIT (sourceforge.net/projects/bpkit):** Open Source (LGPL), Lead designer, 1995-1996. BPKIT was one of the first object-oriented math software packages, and it remains a popular prototyping environment for preconditioned iterative methods.
- **GEMMW (www.mgnet.org/~douglas/ccd-free-software.html):** Open Source (no license), Developer, 1994. GEMMW is a portable parallel implementation of Strassen-Winograd dense matrix-matrix multiplication.
- **Cray Sparse Solvers:** Distributed with Cray Scientific Libraries (LIBSCI), Principal designer and developers of the preconditioned sparse solvers, 1989-1993. ***Provided optimized libraries for sparse linear systems on Cray vector multiprocessor and MPP machines.***
- **Cray Optimized BLAS/LAPACK:** Distributed with LIBSCI, developer of YMP/C90 kernels for vector multiprocessor systems, 1989-1993. ***Developed unique hybrid implementation for single vector processor and multiple vector processors.***
- **Cray vectorized tridiagonal solvers:** Distributed with LIBSCI, principal developer, 1989-1993. ***Developed 3:1 cyclic reduction and burn-at-both-ends algorithms for vector processors.***

Selected Invited Presentations

- ***Better Reproducibility: We do not want it, cannot afford it, but still need it and can have it***, 52nd ORAP Forum: Reproducibility, Paris, France, March 2024.
- ***Accelerating HPC Impact: GPUs, Software Ecosystems, People, Trust***, NASA Seminar Series, April 2022.
- ***Invited: E4S: Toward an Ecosystem for HPC R&D Software***, 2021 Salishan Conference on High Speed Computing, (virtual), April 2021.
- ***Invited Panelist: OSTP Convening: Pioneering the Future Advanced Computing Ecosystem*** (virtual), August 2020.
- ***Keynote: The US Exascale Project Software Stack: Why It Matters to You***, Altair HPC Summit 2020 (virtual), September 2020.
- ***Keynote: The Extreme-scale Scientific Software Stack (E4S) for Collaborative Open Source Software***, Excalibur SLE Conference, Bristol, UK (virtual), July 2020.
- ***Keynote: The US Exascale Project Software Stack: Why It Matters to You***, ANSYS Simulation World Conference 2020 (virtual), June 2020.
- ***Keynote: The Extreme-scale Scientific Software Stack (E4S) for Collaborative Open Source Software***, The 2nd R-CCS International Symposium, Kobe, Japan, February 2020
- ***Keynote: The Extreme-scale Scientific Software Stack (E4S) and Its Promise for the Exascale Era***, The 10th Workshop on Latest Advances in Scalable Algorithms for Large-Scale Systems, ScalA19, SC19, November 2019.
- ***Keynote: About the US Exascale Project and Why ANSYS Might Care***, ANSYS R&D Conference, Pittsburgh, PA, October 2019
- ***Keynote: The Exascale Computing Project's Software Strategy***, HPC User Forum, Lugano, Switzerland, October 2019.
- ***Keynote: Accelerated Sparse Linear Algebra: Emerging Challenges and Capabilities for Numerical Algorithms and Software***, Numerical Algorithms for High Performance Computational Science, London, UK, April 2019.

- **Keynote: *The Extreme-scale Scientific Software Stack (E4S)***, The HPC User Forum, Santa Fe, NM, April 2019.
- **Keynote: *Making Reproducibility Indispensable: Changing the Incentives that Drive Computational Science***, SPPEXA Workshop, Paris, France, March 2019.
- **Keynote: *Accelerated Sparse Linear Algebra: Some Lessons, Challenges and Opportunities***, ISC18, Frankfurt, Germany, June 2018
- **Keynote: *Productive and Sustainable: More Effective CSE***, SIAM Conference on Computational Science and Engineering 2017, Atlanta, GA, February 2017.
- **Keynote: *Strategies for Next Generation HPC Applications and Systems***, ACSI Conference 2016, Fukuoka, Japan, January 2016.
- **Keynote: *A Task-centric/Dataflow Application Architecture for Scalable Systems***, SCALA Workshop 2015, SC'13, Austin, TX, November 2015.
- **Keynote: *Effective use of Miniapps for co-design***. WRAP Workshop, IEEE Cluster, September 2015.
- **Invited: *Efficiency or Productivity: Pick One***. Panelist. SC'14, New Orleans, LA, November 2014.
- **Tutorial: *Scalable Manycore Computing for Sparse Computation***, SC'11, SC'12, SC'13, SC'14.
- **Invited: *Improving Scientific Productivity: Practical Approaches Toward an Elusive Goal***, CEA Invited Talk, Paris, France, October 2014.
- **Keynote: *Productivity for Productivity***, Inter-agency Productivity Workshop, University of Indiana, Bloomington, IN, August 2014.
- **Keynote: *Challenges and Opportunities for Scalable Finite Element Setup & Assembly***, FE Assembly Workshop, Albuquerque, NM, May 2014.
- **Invited: *System Software: A Necessary but Ill-prepared Hero***, Salishan Conference, Salishan, OR, April 2014.
- **Invited: *Toward the Next Generation of Parallel and Resilient Algorithms & Libraries***, Advances in Numerical Algorithms and High Performance Computing, University College London, England, April 2014.
- **Keynote: *Toward the Next Generation of Scalable and Resilient Algorithms***, FP3C Workshop, Maison de Simulation, Paris, France, March 2014.

- **Keynote: *Toward the Next Generation of Parallel and Resilient Algorithms & Applications***, SPPEXA Workshop 2014, Cologne, Germany, December 2013.
- **Keynote: *Toward the Next Generation of Parallel and Resilient Algorithms***, SCALA Workshop 2013, SC'13, Denver, CO, November 2013.
- **Keynote: *Toward Resilient Algorithms and Applications***, FTXS 2013, New York, NY, June 2013.
- **Keynote: *Toward Effective Parallel Programming: What we Need and Don't Need***, HIPS Workshop, IPDPS 2013, Boston, MA, May 2013.
- **Invited: *The Virtues of Data Transparency***, SOS 17, Jekyll Island, SC, March 2013.
- **Invited: *What Every SIAM Member Should Know about Computing on Emerging Architectures***, 2012 SIAM Annual Meeting, Minneapolis, MN, July 2012.
- **Invited: *Scalability of Trilinos: People, Processes, Parallelism***, 2012 ESCO Conference, Pilsen, Czech Republic, June 2011.
- **Invited: *Numerical Libraries on Emerging Architectures***, 2011 Supercomputing Conference Tutorial, Seattle, WA, November 2011.
- **Invited: *Emerging Architectures and UQ: Implications and Opportunities***, IFIP Workshop on uncertainty quantification, Boulder, CO, August 2011.
- **Invited: *Building the Next Generation of Parallel Applications and Libraries***, INT Workshop on Exascale Computing, Seattle, WA, June 2011.
- **Invited: *Toward Portable Programming of numerical linear algebra on manycore nodes***, CEA-EDF-INRIA 2011 Summer School, Nice, France, June 2011.
- **Keynote: *Scalability of Trilinos: People, Processes, Parallelism***, 3rd International Conference on Computational Methods in Engineering and Science (FEMTEC 2011), South Lake Tahoe, NV, May 2011.
- **Invited: *Building the Next Generation of Parallel Applications***, Salishan Conference on High Speed Computing, April 2011.
- **Invited: *Miniapplications: Vehicles for Co-Design***, Engelberg, Switzerland, March 2011.
- **Invited: *Requirements on Next-Generation Programming Models***, U of Houston, January 2011.

- **Invited: *Trilinos for Extreme-scale for Computing***, U Texas, Austin, January 2011.
- **Invited: *Software Engineering for Computational Science and Engineering***, Cray, Inc., January 2011.
- **Invited: *Building the Next Generation of Parallel Applications and Libraries***, IAM, April 2011.
- **Invited: *Bi-modal MPI-only & MPI+threading***, Cray, Inc., December 2010.
- **Invited: *The Extreme-scale Algorithms & Software Institute***, Fall Creek Falls Conference, October 2010, Memphis, TN.
- **Invited: *Building the Next Generation of Scalable Applications***, Future of the Field Workshop, Snowbird, UT, July 2010.
- **Keynote: *Building the Next Generation of Parallel Applications***, Int'l Workshop on OpenMP, Tsukuba, Japan, June 2010.
- **Keynote: *Trilinos for Extreme-scale Computing***, SPEEDUP Workshop, ETH-Zurich, September 2010.
- **Invited: *Trilinos Overview and Tutorial***, Purdue University, September 2009.
- **Invited: *Software Needs for Next-generation systems***, SOS13, Hilton Head, SC, March 2009.
- **Invited: *Algorithms for 1M cores: What Might and Might not Work***, Simulating the Future Workshop, Paris, France, September 2008.
- **Organizer: *When MPI-only is not Enough: Building the Next Generation of Scalable Applications Workshop***, Santa Fe, NM, May 2008.

Research Funding

Year	Source	Role	Amount	Summary
2018-2023	DOE ECP	Director	\$510M	ECP Software Technology Portfolio
2016-2017	DOE ECP	Lead	\$15M	ECP Math Libraries Portfolio
2016-2017	DOE ECP	PI	\$5M	xSDK Math Libraries
2016-2017	DOE ECP	PI	\$5M	IDEAS Scientific Productivity
2016-2017	DOE ECP	PI	\$500K	ForTrilinos – Fortran APIs to Trilinos
2016-2017	DOE ECP	PI	\$400K	PEEKS Scalable Solvers
2015-2017	DOE ASCR/BER	PI	\$12M	IDEAS Productivity
2013-2015	DOE-NNSA Co-Design	Co-PI	\$600K	ASC Co-Design Efforts.
2010-2012	DOE-NNSA Exascale Alg	PI	\$600K	Exascale Co-Design Initiative
2010-2012	ASC/Algs	Co-PI	\$1.2M	Technical Lead
2010-2011	DOE-SC ASCR	PI	\$168K	Extreme-scale Software Center (ESC)
2010-2012	DOE-SC ASCR	PI	\$3M	Extreme-scale Alg & SW Institute (EASI)
2010-2012	ASC/CSSE	Co-PI	\$1M	Mantevo Miniapps.
2010-2011	DOE-SC ASCR	PI	\$1.1M	SciDAC-e Solvers for Carbon Sequestration.
2010-2012	DARPA	Co-PI	\$3.5M	UHPC Xcaliber Project.
2009-2012	DOE-SC ASCR	PI	\$1.2M	Applied Math Base Program (Tramonto).
2008-2010	DOE-SC ASCR	PI	\$3.5M	Institute for Advanced Architectures & Algorithms.
2007-2009	LDRD	PI	\$1.1M	Mantevo Miniapps initiation.
2006-2011	DOE-SC ASCR	PI	\$1M	SciDAC TOPS2 project.

2001-2003	LDRD	Co-PI	\$1M	Data Partitioning Tools.
2000-2010	ASC/Algs	PI, Co-PI	\$1.2M	ASC Algorithms leadership
1999-2002	LDRD	PI	\$1M	Hybrid Preconditioners for coupled systems.

Mentoring Highlights

- **Current Mentoring:**
 - Connor Brynteson, St. John's University – Use of generative AI tools for advance software product development
 - Evan Eisinger, St. John's University – Characterizing Research Software Science relative to meta-science, use of generative AI tools for upstream software product development activities
- **Past Sandia staff:** Siva Rajamanickam, Elaine Raybourn, Ross Bartlett, Brian Reuters, Wesley Coomer, Richard Barrett, Ross Bartlett, Erik Boman, Russell Hooper, Jonathan Hu, Nicole Lemaster, Mike Parks, Michael Wolf, Roger Pawlowski, Denis Ridzal, Chris Siefert, Heidi Thornquist, Andy Salinger, Guglielmo Scovazzi: Informal staff mentoring.
- **Carter Grove, Fiona Smith, Amiah Brower:** Undergraduate research advisor, 2021 – 2024.
- **Irina Demeshko:** Postdoctoral advisor, 2015 – 2017.
- **Andrey Prokopenko:** Postdoctoral advisor, 2015 – 2017.
- **Alicia Klinvex:** Postdoctoral advisor, 2014 – 2016. Graduate mentor, 2014 – 15.
Advisor: Ahmed Sameh, Purdue University, graduating May 2015.
- **Grey Ballard:** Postdoctoral advisor, 2013 – 2015.
Advisor: Jim Demmel, UC-Berkeley.
- **Chris Baker:** Postdoctoral advisor, 2013 – 2015.
- **Mawussi Zounon:** PhD Committee member, University of Bordeaux, 2015.
- **France Boillod-Cerneax:** PhD Committee member, University of Lille, Paris, 2014 – 2015.
- **Emily Furst:** Undergraduate advisor, St. John's University, 2012 – 2015.
- **Zachary Bookey:** Undergraduate advisor, St. John's University, 2013 – 2015.
- **Alyssa Anderson:** Undergraduate research advisor, 2012 – 2014.
- **Christian Trott:** Postdoctoral advisor, 2012 – 2014.
- **Ben Seefeldt:** Undergraduate research advisor, 2011 – 2013.
- **Jacob Hemstad:** Undergraduate research advisor, 2011 – 2013.

- **Brandon Hildreth:** Undergraduate research advisor, 2011 – 2013.
- **Mark Hoemmen:** Postdoctoral advisor, 2009-2012.
Advisor: Jim Demmel, U of CA, Berkeley.
- **Siva Rajamanickam:** Postdoctoral advisor, 2009-2012.
Advisor: Tim Davis, U of FL.
- **Michael Wolf:** Postdoctoral advisor, 2009-2011.
Advisor: Mike Heath, U of IL, Urbana-Champaign.
- **Sarah Knepper:** Undergraduate research advisor, summer intern advisor, PhD committee, 2004-2011.
Advisor: Jim Nagy, Emory University.
- **Kurtis Nusbaum:** Undergraduate research advisor, Sandia intern mentor, 2008-2011.
- **Kelsey Larson:** Undergraduate research advisor, 2010 – 2012.
- **Becca Simon:** Undergraduate research advisor, 2010 – 2012.
- **Lee Buermann:** Undergraduate research advisor, 2010 – 2012.
- **Christopher Baker:** Postdoctoral advisor, 2008-2009.
Advisor: Kyle Gallivan, U of FL.
- **Matthew Lietzke:** Undergraduate research advisor, 2007-2008.
- **Eric Bavier:** Undergraduate research advisor, 2008-2010.
- **Vanja Paunic:** Undergraduate research advisor, 2007-2008.
- **Kelly Fermoye:** Undergraduate research advisor, summer intern mentor, 2006-2010.
- **Michael Karp:** Undergraduate research advisor, 2006-2007.
- **Jason Cross:** Undergraduate research advisor, summer intern mentor Sandia, 2002-2005.
- **James Willenbring:** Undergraduate Research Advisor, St. John's, 2000-2001.
- **Kristopher Kampshoff:** Undergraduate Research Advisor, St. John's, 1999-2001.
- **Abdelkader Baggag:** Summer Research at Cray, 1997.
Advisor: Ahmed Sameh, U of Purdue.

- **Edmond Chow:** Summer Research at Cray, 1996-1997.
Advisor: Yousef Saad, U of Minnesota.
- **John Wu:** Summer Research at Cray, 1994-1995.
Advisor: Yousef Saad, U of Minnesota.
- **Keith Gremban:** Thesis committee, summer research, 1992-1995.
Advisor: Gary Miller, Carnegie-Mellon.
- **Vivek Sarin:** Summer Research at Cray, 1992-1994.
Advisor: Ahmed Sameh, U of Minnesota.